A Small Station-Graph World Model for 0–24 h Weather in North India: A Version-by-Version Record Against Global AI Weather Models and Time-Series Foundation Models

Aditya Kumar
ParadoxAI Lab
India
contact.paradoxailab@gmail.com

Anupam Singh
ParadoxAI Lab
India

ABSTRACT

Global AI weather models such as ECMWF's AIFS now match or beat physics-based forecasts on grid averages, and zero-shot time-series foundation models such as Google's TimesFM and Amazon's Chronos forecast any series without training. Neither is built for what many users in India need: the next 24 hours of temperature, humidity and wind at a specific station. We study whether a small model trained only on a station network's own history can do better. PWM-W is a graph world model of about one million parameters in which every station is a token. It comes in two forms: a pure form that uses observations only, and a hybrid that also reads global forecasts published before the forecast starts. We test it in three north-Indian leagues: six airports (METAR), 20 NOAA stations around Delhi (WMO) and 25 Delhi air-quality stations' weather sensors (CPCB). The opponents are real archived operational forecasts of AIFS, IFS, ICON, GFS and NOAA's GraphCast-based AIGFS, scored raw and after a trailing local bias correction, together with TimesFM 2.5 and 3.0 and Chronos-2. Every change from version 0.3 to 0.9 was adopted only if it beat the default setting on validation by more than two seed standard deviations, and several proposed changes failed that test. At the airports, the hybrid beats every global model on every variable it forecasts when each is scored on its newest run published before the forecast starts (AIFS by 4.4%, 11.8% and 7.4%). Against AIFS's freshest run it still loses on temperature and humidity. The pure model beats all three foundation models on every variable at the airport and WMO stations. On the noisier CPCB sensors, cross-network training, a 14-day encoder and weight averaging turned four of its losses into ties, but it still trails Chronos-2 on temperature and all three foundation models on wind. Bias-corrected FourCastNet and GenCast Mini beat the pure model on airport temperature. We report every result, including the losses, and the test-reuse risk of a version-by-version study. A data-threshold study shows that PWM-W needs 12–24 months of a network's own history to stop losing to the foundation models and 24–48 months to beat all three on every variable.

KEYWORDS

weather forecasting, station forecasts, post-processing, graph neural networks, time-series foundation models, AI weather models, India, benchmarking

1 INTRODUCTION

Machine-learned global weather models now forecast the atmosphere at least as well as physics-based systems on standard grid-level scores [Lam et al. 2023; Lang et al. 2024; Price et al. 2025]. ECMWF runs its AI model, AIFS, operationally, and NOAA runs a GraphCast-based system (AIGFS). In parallel, time-series foundation models such as TimesFM [Das et al. 2024] and Chronos [Ansari et al. 2025, 2024] forecast any series zero-shot from its own history.

Many decisions in India depend on a narrower question: what will the thermometer, hygrometer and anemometer at this airport or this monitoring station read over the next day? Global models answer it only through a grid cell of roughly 9–28 km, which misses local siting and land use. Foundation models see one sensor at a time and no weather physics. The classical answer is statistical post-processing of numerical forecasts at stations [Glahn and Lowry 1972; Gneiting et al. 2005; Rasp and Lerch 2018].

We ask how far a small model can go that is trained only on a station network's own history, optionally with the global forecasts that exist when the forecast is made. Our model, PWM-W, has about one million parameters, more than 100 times fewer than Chronos-2 (120M) and TimesFM 2.5 (200M). We report its development version by version under a fixed rule, including the changes that failed.

Contributions.

- **Three north-Indian station leagues** (§3): airports, NOAA's quality-controlled stations around Delhi, and Delhi's air-quality stations' own weather sensors. The opponents are real archived operational forecasts of five global systems. Each is scored under two rules: its *latest run*, which is often issued after our forecast starts, and its *newest run issued before* our forecast starts. Each is also scored raw and after a trailing local bias correction. TimesFM 2.5, TimesFM 3.0 and Chronos-2 complete the field.
- **PWM-W** (§4), an inductive station-graph world model, in a pure form (observations only) and a hybrid form that also reads bias-corrected forecasts from four global models, either as inputs or as a base it learns to correct.
- **A version-by-version record** (§6) from v0.3 to v0.9. Each change was adopted only if it beat the default setting on validation by more than two seed standard deviations. Several proposals failed this test, and we report them alongside the ones that passed.
- **Results with losses stated** (§7):
  - at the airports, the hybrid beats AIFS, IFS, ICON, AIGFS and GFS on every variable they forecast under the issued-before rule, but loses to AIFS's freshest run on temperature and humidity;
  - the pure model beats all three foundation models at the airport and WMO stations;

Table 1: The three leagues. Origins: test forecasts, one every 6 h.

	METAR	WMO	CPCB
Source	IEM (METAR)	NOAA ISD	CPCB/DPCC/IMD
Stations	6 airports	20	25
Clock	UTC	UTC	IST
Train	2021-01–2025-10	2021-01–2024-11	2024-01–2025-05-02
Validation	2025-11–2026-01	2024-12–2025-02	2025-05-03–07-02
Test	2026-02–2026-10	2025-03–2025-08-20	2025-07-03–12-31
Test origins	1,089	689	725

  - on the CPCB sensors the pure model still trails the foundation models on wind, while the hybrid ties or beats them;
  - bias-corrected FourCastNet and GenCast Mini, which we ran ourselves, beat the pure model on temperature.
- **A data threshold and a real-time track** (§8): how many months of local history PWM-W needs to match or beat the foundation models, and a live forecast logged before the weather happens.

2 RELATED WORK

Global AI weather models. These include GraphCast [Lam et al. 2023], Pangu-Weather [Bi et al. 2023], FourCastNet [Pathak et al. 2022] and its spherical successor SFNO [Bonev et al. 2023], the diffusion ensemble GenCast [Price et al. 2025], ECMWF's operational AIFS [Lang et al. 2024] and Aurora [Bodnar et al. 2025]. All learn from reanalysis on global grids of 0.25° or coarser, and they are evaluated mainly on grid-level scores [Rasp et al. 2024]. We use their real operational output at stations, and we run FourCastNet SFNO and GenCast Mini ourselves.

Station post-processing. Model output statistics [Glahn and Lowry 1972] and ensemble MOS [Gneiting et al. 2005] correct numerical forecasts at stations. Neural post-processing improves on them [Rasp and Lerch 2018]. Our hybrid belongs to this family. Its differences are that it is a graph model over the whole station network, and that it is trained jointly with a pure observation-only forecaster that shares its architecture.

Time-series foundation models. TimesFM [Das et al. 2024], Chronos [Ansari et al. 2024], Chronos-2 [Ansari et al. 2025], Moirai [Woo et al. 2024] and Sundial [Liu et al. 2025] forecast unseen series zero-shot. They are strong baselines for any single sensor, but they see neither the station network nor the weather physics.

Spatio-temporal graph models. DCRNN [Li et al. 2018], Graph WaveNet [Wu et al. 2019] and STID [Shao et al. 2022] model sensor networks. PWM-W follows the inductive line, in which no per-sensor embedding is used, so the same weights run on networks of any size.

3 LEAGUES AND DATA

Every league forecasts 2 m temperature (°C), relative humidity (%) and 10 m wind speed (m/s) for the next 24 hours. Forecast origins are every 6 hours, and only observations up to the hour before the origin are used. Stations and splits were fixed by rules written before any score was seen (Table 1).

METAR (airports). These are routine airport reports from the Iowa Environmental Mesonet archive. The candidates were every Indian station within 600 km of Delhi Palam and below 1,000 m, of which at most 16 could be kept, and a station was kept only if at least 60% of its hours had a temperature in both periods. Six airports qualified: Delhi Palam, Jaipur, Amritsar, Lucknow, Udaipur and Bhopal. Each hourly value is the report nearest the top of the hour, within ±15 min.

WMO (NOAA ISD).. These are quality-controlled synoptic and METAR records from NOAA's Integrated Surface Database [Smith et al. 2011]. We took the nearest stations within 500 km of Delhi and below 1,000 m with at least 10% coverage, up to 24. Of these 24, 20 passed the model's coverage rule, including Delhi Safdarjung and Palam. Many report only every 3 hours, so many hours are missing, and scores use observed hours only. NOAA's Indian records end on 2025-08-24, which fixes the test period.

CPCB (Delhi air-quality stations). These are the weather sensors at Delhi's air-quality monitoring stations. Stations were located by name with OpenStreetMap, and 25 of 37 passed the coverage rule. Runs of identical readings of 12 h or longer are masked as stuck. The sensors are not sited to WMO standards: many are on rooftops, and wind is often sheltered (19.6% of wind hours are missing). The league therefore tests forecasts of what these sensors will read, which matters for air-quality work. It is not a test of free-air weather.

Archived operational forecasts. Open-Meteo's Previous Runs API [Zippenfenig 2023] stores the forecasts that operational models actually issued. It gives each model's latest run and its runs from 1 and 2 days earlier, which we read at each station's nearest 0.05° point. The models are ECMWF AIFS [Lang et al. 2024], ECMWF IFS at 9 km and at 0.25°, DWD ICON, NOAA GFS and NOAA AIGFS (GraphCast-based). AIGFS has a short archive and no humidity, so it appears in the METAR league only.

4 THE PWM-W MODEL

Station tokens. Each station becomes one token, built by a two-layer MLP from:

- its last 72 hours of temperature, humidity and wind, with observed flags;
- its last 24 hours of the other channels (wind direction, dew point or solar radiation, pressure, visibility or $PM_{2.5}$, depending on the league);
- its mean daily cycle over the last 14 days and its trailing 7-day level;
- the sine of the sun's elevation at the station for each of the 24 forecast hours;
- its position, at a fixed scale that is the same in every network, and the day of year.

Hour-of-day and month embeddings are added. There is no per-station embedding, so the same weights run on any station network.

Mixing across stations. Three blocks of multi-scale graph attention mix the tokens. Each head's attention bias is a learned mixture of the k-nearest-neighbour graph, its square and the identity, so different heads specialise in different ranges. Region tokens, pooled

over spectral clusters, and a global token carry information between distant stations. The model has width 128 and 4 heads.

Output. A linear head gives the 10%, 50% and 90% quantiles for each variable and each of the 24 hours. They are expressed as offsets, in units of the network's spread, from a base: the recent daily cycle plus the latest anomaly. Training minimises the quantile (pinball) loss [Koenker and Bassett 1978]. Training randomly darkens stations and blanks bursts of hours before the origin, so that the model learns to handle real outages. The reported forecast is the mean of three seeds. Intervals are calibrated on validation by conformalised quantile regression [Romano et al. 2019], but this paper scores only the median.

Hybrid. The hybrid also reads global forecasts, using for each model the newest run issued before the forecast starts: the 1-day-old run for hours 0–16 and the 2-day-old run after. Each forecast is bias-corrected by its trailing 30-day error at that station and hour of day. Two hybrid designs compete on validation:

- **H1** adds the corrected forecasts of IFS 0.25, AIFS, ICON and GFS as inputs;
- **H2** replaces the base with their mean, a model-output-statistics base, and learns only the correction.

Additions in later versions (§6).

- **Cross-network training (v0.6).** Half of the training batches come from another station network. It uses only that network's hours before this league's validation period, with its own normalisation statistics from those hours alone. Other channels are matched by name, and the clock is shifted to the league's.
- **14-day encoder (v0.8).** The full 14×24-hour history, previously only averaged into a daily cycle, passes day by day through a small MLP with attention pooling over days.
- **Weight averaging (v0.8).** An exponential moving average of the weights (decay 0.998) is what gets validated and used.

The final models have 0.93–1.13M parameters.

5 PROTOCOL

Opponents.

- **Global models** (§3), each scored on four versions: raw and bias-corrected, under each of the two run rules below. The bias correction is the trailing 30-day mean error at that station and hour of day, taken from days before the valid time, with at least 10 such days required.
- **The bias-corrected mean of IFS 0.25, AIFS, ICON and GFS**, issued before the forecast starts: a classical model-output-statistics baseline.
- **TimesFM 2.5** (200M parameters), **TimesFM 3.0** and **Chronos-2** (120M), zero-shot on each station's last 512 hours. TimesFM 3.0 is used under its non-commercial research licence.
- **Three simple baselines**: the same hour yesterday, the 14-day daily cycle, and the daily cycle plus the latest anomaly.

Two rules for global models.

- **STRICT** scores a global model on its latest run. That run is often issued after our forecast starts, which favours the global model.
- **ISSUED-BEFORE** scores it on its newest run published before our forecast starts: the 1-day-old run for hours 0–16 and the 2-day-old run after. This is what a forecaster could actually have used.

Under either rule, PWM-W *beats* a global model on a variable only if its mean absolute error (MAE) is significantly lower than both the raw and the bias-corrected forecast.

Significance. A difference counts only if the Diebold–Mariano test [Diebold and Mariano 1995] gives $p < 0.05$ and a day-block bootstrap [Künsch 1989] 95% interval excludes zero. Otherwise we call it a tie. Each comparison uses only the hours where both forecasts and the observation exist.

Model choice on validation only. Three seeds of the default setting, P0, give a seed standard deviation σ. A candidate replaces P0 only if its validation score is lower by more than 2σ. The validation score is MAE in units of each variable's network spread, averaged over the three variables. The best passing candidate is retrained with three fresh seeds for the test. The same rule decides between hybrid designs. Appendix A lists every candidate.

6 VERSION-BY-VERSION RECORD

Table 2 lists every change from v0.3 to v0.9, the validation evidence and the decision. The validation numbers are the score of §5, where lower is better. Changes that failed the rule are reported in the same way as changes that passed.

What each version changed in practice.

- **v0.3** set up the leagues. It compared global models only under STRICT, which our own analysis showed to favour them, because their latest run often postdates our forecast start.
- **v0.4** added the ISSUED-BEFORE rule and the learned-correction hybrid. At the airports this moved the hybrid's MAE from 1.170 to 1.008 °C (temperature), 5.94 to 5.49% (humidity) and 0.962 to 0.889 m/s (wind). Cross-network training between the airports and the WMO stations did not pass the rule, and was not used there.
- **v0.5** added more predictors from the global runs. It failed in both leagues where it was tried. A fresh retraining of v0.4's design reproduced v0.4's test numbers exactly, which confirms that the pipeline is deterministic.
- **v0.6** targeted the CPCB league, the one league where the pure model lost to the foundation models. That league has 16 months of training data. Cross-network training with the WMO stations improved validation by about 5σ. The same version fixed a bug that had invalidated v0.4's TimesFM 2.5 wind comparison on CPCB: after running out of GPU memory on wind, the runner had reused the humidity forecast. We withdrew that number.
- **v0.7** tested weekly retraining. Simulated on the validation period, it made the forecast worse, and it was rejected.
- **v0.8** let the model read the 14 days it already received hour by hour instead of only as an average, and averaged its weights. On one seed, each change alone improved on P4 by less than

Table 2: Every change, the validation evidence, and the decision. Val: validation MAE in units of network spread (lower is better); ±: seed standard deviation of P0. Test effects are reported in §7.

Version	Change	Validation evidence	Decision
v0.3	First release: three leagues, pure model with candidates P1 (wider), P2 (no graph), P3 (sparse experts); hybrid H1 with IFS 0.25 and AIFS runs at least 48 h old; STRICT rule only	WMO: removing the graph is worse (0.3948 vs 0.3864)	P0 kept
v0.4	ISSUED-BEFORE rule; four models in the hybrid; learned-correction hybrid H2; bias-corrected four-model ensemble added as an opponent; cross-network training P4 (airports ↔ WMO); fixed position scaling	H2 vs H1: METAR 0.348 vs 0.372; WMO 0.3732 vs 0.3783; CPCB 0.4710 vs 0.4540. P4: METAR 0.3920 vs P0 0.3934 ± 0.0012; WMO 0.3889 vs 0.3879 ± 0.0010. CPCB: P1 0.4797 vs P0 0.4922 ± 0.0038 (v0.6 rerun)	H2 (METAR, WMO) H1 (CPCB) P4 rejected P1 (CPCB)
v0.5	Hybrid H3: H2 plus cloud cover, solar radiation, dew point, pressure and wind direction from the same runs	METAR 0.3539 vs H2 0.3481; CPCB (in the v0.6 run) 0.4932 vs H1 0.4586	rejected
v0.6	Cross-network training for CPCB, with the 20 WMO stations as partner (hours before CPCB validation only); fix to the foundation-model runner	CPCB: P4 0.4735 vs P0 0.4922 ± 0.0038 (P1 0.4797)	P4 (CPCB)
v0.7	Weekly rolling fine-tuning on hours observed before each week, simulated on validation	CPCB: rolling 0.4727 vs frozen 0.4650	rejected
v0.8	14-day encoder (P6), weight averaging (P7), both (P8), all on top of P4	CPCB: P6 0.4714, P7 0.4721, P8 0.4678 vs P0 0.4922 ± 0.0038	P8 (CPCB)
v0.9	21-day patch encoder PWM-X (P9), plus masked-patch reconstruction (P10), on P4 with weight averaging; data-threshold study (§8)	CPCB: P9 0.4606, P10 0.4597 vs P8 0.4678, P0 0.4922 ± 0.0038	P10 (CPCB)

1σ; together (P8) they scored 6.4σ better than P0, most of which is P4's own gain. P8's margin over v0.6's P4 (0.0057) is smaller than 2σ, as was P4's margin over v0.4's P1 (0.0062). The rule compares every candidate with P0, so its preference among passing candidates is not decisive.

Versions 0.6–0.9 were run on CPCB only (the threshold study also covers the airports). The airport and WMO results below use v0.4's chosen designs: P0 for the pure model and H2 for the hybrid.

7 RESULTS

Differences are relative MAE: negative means PWM-W is better. Every non-tie verdict passed both significance tests.

7.1 Airports (METAR, v0.4 design)

Under ISSUED-BEFORE, the hybrid beats every global model on every variable it forecasts (Table 4). Under STRICT, against each model's freshest run, the picture is mixed:

- **Wind:** the hybrid still beats all six models.
- **Humidity:** it beats IFS and ICON and loses to AIFS (+3.3%).
- **Temperature:** it loses to AIFS (+15.6%) and IFS 0.25 (+3.3%), ties IFS 9 km and ICON, and beats AIGFS and GFS.

The pure model, with no global input, beats all three foundation models on every variable. Against the bias-corrected four-model ensemble it loses on temperature (+16.6%) and wind (+10.2%) and ties on humidity: at airports, with good global forecasts available, the hybrid is the model to use.

Table 3: Airports, temperature MAE (°C), test 2026-02 to 2026-10. BC: trailing 30-day bias correction. Latest: the run used under STRICT. Issued-before: the newest run published before the forecast starts.

Entry	MAE
ECMWF AIFS, latest run + BC	0.872
ECMWF IFS 0.25, latest run + BC	0.976
ECMWF IFS 9 km, latest run + BC	0.986
DWD ICON, latest run + BC	1.005
PWM-W hybrid	**1.008**
ECMWF AIFS, issued-before + BC	1.054
Four-model ensemble, issued-before + BC	1.083
PWM-W pure	**1.263**
TimesFM 3.0	1.341
Chronos-2	1.357
TimesFM 2.5	1.403
NOAA GFS, latest run, raw	2.296

7.2 WMO stations (v0.4 design)

- **Pure model, ISSUED-BEFORE:**
  - it beats IFS 0.25 on all three variables (−3.3%, −15.7%, −22.2%), ICON on all three, and GFS;
  - it beats AIFS on humidity (−2.8%) and wind (−18.8%) but loses on temperature (+14.1%);
  - it beats TimesFM 2.5, TimesFM 3.0 and Chronos-2 on every variable ($p < 10^{-5}$).

Table 4: Airports: the hybrid against each global model under ISSUED-BEFORE (vs. the bias-corrected run); all BEATS. Bottom rows: the pure model against the foundation models (all wins, $p < 0.01$).

Opponent	Temp	RH	Wind
ECMWF AIFS	−4.4%	−11.8%	−7.4%
ECMWF IFS 9 km	−21.0%	−24.2%	−13.4%
ECMWF IFS 0.25	−19.4%	−22.0%	−12.7%
DWD ICON	−22.1%	−23.6%	−11.0%
NOAA AIGFS	−21.0%	–	−17.7%
NOAA GFS	−37.2%	−41.4%	−29.3%
Four-model ensemble	−6.9%	−11.5%	−3.6%
Pure model vs. foundation models			
TimesFM 3.0	−5.8%	−4.3%	−1.5%
TimesFM 2.5	−10.0%	−7.3%	−2.4%
Chronos-2	−6.9%	−5.6%	−1.8%

Table 5: CPCB: the pure model (v0.6) against each global model under ISSUED-BEFORE, vs. the bias-corrected run. All BEATS. Under STRICT every cell is also BEATS except humidity vs. AIFS (tie, −5.0%).

Opponent	Temp	RH	Wind
ECMWF AIFS	−27.7%	−7.1%	−50.2%
ECMWF IFS 9 km	−49.3%	−36.8%	−56.2%
ECMWF IFS 0.25	−33.6%	−25.2%	−45.1%
DWD ICON	−33.3%	−22.4%	−34.9%
NOAA GFS	−46.3%	−45.3%	−55.9%

Table 6: CPCB: the pure model against the foundation models, by version. Positive values mean the foundation model is better. W/L: significant win/loss; T: tie. v0.4's TimesFM 2.5 wind comparison was invalid (§6).

Opponent	Version	Temp	RH	Wind
Chronos-2	v0.4	+8.3 L	+6.9 L	+6.0 L
	v0.6	+6.8 L	+3.7 L	+4.9 L
	v0.8	+5.0 L	+1.4 T	+3.8 L
TimesFM 3.0	v0.4	+5.7 L	+6.8 L	+6.7 L
	v0.6	+4.3 L	+3.6 L	+5.5 L
	v0.8	+2.5 T	+1.3 T	+4.4 L
TimesFM 2.5	v0.4	+2.1 T	+4.1 L	invalid
	v0.6	+0.7 T	+0.9 T	+4.6 L
	v0.8	−1.0 T	−1.3 T	+3.5 L

- **Hybrid:** it beats IFS, ICON and GFS on all variables, beats AIFS on humidity and wind, and ties AIFS on temperature (−1.7%). It beats the four-model ensemble by 2.4%, 4.9% and 16.6%.

7.3 Delhi air-quality station sensors (CPCB)

- **Against the global models** (Table 5), the pure model wins by large margins, because a global model with a mean bias correction cannot learn the quirks of rooftop sensors. In v0.8 the pure model beats the bias-corrected four-model ensemble by 28.8%, 13.2% and 38.5%.
- **Against the foundation models** (Table 6), v0.6 and v0.8 narrowed every gap, and three losses became ties. Wind is still lost to all three.
- **The v0.8 hybrid** ties all three foundation models on temperature and humidity. It beats Chronos-2 (−3.3%, $p = 0.03$) and TimesFM 2.5 (−3.6%, $p = 0.02$) on wind and ties TimesFM 3.0 (−2.7%). These two wind wins would not survive a correction for the nine hybrid-versus-foundation-model tests on this league (§9).

Table 7: Airports at +6/12/18 h from 00 UTC: the pure model against models we ran. MAE; BC: trailing 30-day bias correction from the model's own earlier days. The hybrid has not yet been scored here.

Model (days)	Var	PWM-W	Raw	BC	Verdict
FourCastNet SFNO (273)	Temp	1.446	1.832	1.278	loses
	Wind	1.055	1.204	1.052	tie
GenCast Mini (88)	Temp	1.813	2.953	1.202	loses
	Wind	1.036	1.219	1.067	tie

7.4 Models we ran ourselves (airports)

We ran NVIDIA Earth2Studio models from the real GFS analysis at 00 UTC on each test day, and read each station at +6, +12 and +18 h (Table 7). FourCastNet SFNO completed all 273 days, and GenCast Mini (1°, one member) completed 88. GraphCast at 0.25°, FourCastNet 3 and Aurora did not fit in the 15 GB memory of the free T4 GPU. GraphCast Small ran but returned only missing values, probably because it needs a precipitation input that GFS analyses lack; we did not verify this.

The pure model beats both raw forecasts. Once their local bias is removed, both beat it on temperature, and wind is a tie.

8 VERSION 0.9: HOW MUCH DATA, AND A NEW ENCODER

8.1 Data threshold

How much of a network's own history does PWM-W need to match the foundation models, which use no training data? For each budget of M months, we train PWM-W on only the last M months before the validation period. The training windows, their inputs and every normalisation statistic all lie inside the budget, and no partner network is used. The setting is fixed in advance: P8 without cross-network training (width 128, 3 layers, 14-day encoder, weight averaging), the mean of 3 seeds. The fixed validation period picks only the training checkpoint, so "M months" means M months of training data plus that period. Stations, test windows and significance tests are the league's.

The answer is clear at the airports (Table 8). With 12 months of history PWM-W mostly ties. With 24 months it loses nowhere. With 48 months it beats all three foundation models on every variable. The WMO stations, with more stations and more varied sites, get there faster: no losses at 12 months and wins everywhere at 24. On

Table 8: Data threshold: pure PWM-W trained on the last M months only, against each foundation model (relative MAE; positive means the foundation model is better). W/L: significant win/loss; T: tie. CPCB has 16.3 months of training data in total; the airports have 58.8 and the WMO stations 47.7.

		Chronos-2			TimesFM 2.5			TimesFM 3.0		
League	M	Temp	RH	Wind	Temp	RH	Wind	Temp	RH	Wind
CPCB	1	+61.3 L	+70.5 L	+36.6 L	+52.1 L	+66.0 L	+36.1 L	+57.5 L	+70.4 L	+37.4 L
	2	+50.9 L	+36.8 L	+22.9 L	+42.3 L	+33.2 L	+22.5 L	+47.4 L	+36.7 L	+23.6 L
	3	+39.5 L	+30.2 L	+19.3 L	+31.5 L	+26.7 L	+18.9 L	+36.2 L	+30.0 L	+20.0 L
	6	+19.1 L	+14.8 L	+15.7 L	+12.3 L	+11.8 L	+15.3 L	+16.3 L	+14.7 L	+16.4 L
	12	+10.2 L	+7.6 L	+6.1 L	+3.9 L	+4.8 L	+5.8 L	+7.6 L	+7.5 L	+6.8 L
	16.3	+7.9 L	+6.4 L	+4.9 L	+1.7 T	+3.6 L	+4.5 L	+5.3 L	+6.3 L	+5.5 L
Airports	1	+46.1 L	+63.7 L	+23.1 L	+41.2 L	+60.7 L	+22.3 L	+47.8 L	+66.0 L	+23.6 L
	2	+12.0 L	+17.3 L	+11.5 L	+8.3 L	+15.2 L	+10.8 L	+13.3 L	+18.9 L	+11.9 L
	3	+11.2 L	+16.1 L	+10.6 L	+7.5 L	+14.0 L	+9.9 L	+12.5 L	+17.8 L	+11.1 L
	6	+5.9 L	+6.3 L	+6.2 L	+2.4 L	+4.3 L	+5.5 L	+7.2 L	+7.7 L	+6.6 L
	12	+0.9 T	+1.2 T	+2.1 L	−2.5 W	−0.6 T	+1.5 L	+2.1 L	+2.7 L	+2.5 L
	24	−1.9 W	−1.3 T	−0.2 T	−5.2 W	−3.0 W	−0.8 T	−0.8 T	+0.1 T	+0.2 T
	48	−6.3 W	−5.2 W	−2.1 W	−9.4 W	−6.9 W	−2.7 W	−5.2 W	−3.9 W	−1.7 W
	58.8	−7.4 W	−5.9 W	−2.6 W	−10.5 W	−7.6 W	−3.3 W	−6.3 W	−4.5 W	−2.3 W
WMO	1	+37.3 L	+24.1 L	+16.4 L	+31.6 L	+19.9 L	+16.2 L	+37.7 L	+24.7 L	+17.3 L
	2	+14.2 L	+9.5 L	+9.9 L	+9.5 L	+5.8 L	+9.6 L	+14.6 L	+9.9 L	+10.6 L
	3	+10.6 L	+7.1 L	+7.2 L	+6.0 L	+3.4 L	+7.0 L	+10.9 L	+7.5 L	+8.0 L
	6	+3.4 L	+7.2 L	+3.1 L	−0.8 T	+3.6 L	+2.9 L	+3.8 L	+7.7 L	+3.8 L
	12	−3.5 W	−1.9 W	−0.4 T	−7.4 W	−5.3 W	−0.6 T	−3.1 W	−1.5 W	+0.3 T
	24	−8.7 W	−5.6 W	−2.2 W	−12.4 W	−8.8 W	−2.4 W	−8.4 W	−5.2 W	−1.5 W
	47.7	−11.3 W	−8.1 W	−3.4 W	−14.9 W	−11.3 W	−3.6 W	−11.0 W	−7.8 W	−2.7 W

the CPCB sensors, 16 months of local history are not enough: training on the network alone, it still loses on almost every comparison. Cross-network training (v0.6) and the later encoders recover much of that gap, as shown below. At the airports, the step from 12 to 48 months moved every comparison by 4–7 percentage points; on CPCB, PWM-X's encoder moved the same comparisons by 0.6–1.6 points.

8.2 PWM-X: a 21-day patch encoder

PWM-X adds a second encoder to each station token. It reads the station's last 21 days as 21 daily patches of the three targets and their observed flags. Each window is normalised by its own masked mean and standard deviation, which is how the foundation models see a series. A two-layer transformer encodes the patches, and attention pooling plus the last patch feed a zero-initialised projection added to the token. P10 also hides 15% of the patches during training and reconstructs them, a self-supervised loss with weight 0.1. P9 has 1,251,709 parameters and P10 1,260,997. Both run on top of P4 (cross-network training) with weight averaging.

On CPCB validation, P9 scored 0.4606 and P10 0.4597, against P8's 0.4678 and P0's 0.4922 ± 0.0038; the rule chose P10. Its margin over P8 (0.0081) only just exceeds 2σ (0.0076). The hybrid on P10 chose H1 (validation 0.4506; H2 0.4607, H3 0.4958), slightly worse than v0.8's hybrid (0.4444).

PWM-X narrowed all nine gaps (Table 9) but changed no verdict: wind still loses to all three, and temperature to Chronos-2. The pure model beats every global model on every variable under both rules, and the bias-corrected four-model ensemble by 29.9%, 14.2% and 38.9%. The hybrid on PWM-X ties all three foundation models on temperature and humidity. It beats Chronos-2 (−3.0%, $p = 0.046$) and TimesFM 2.5 (−3.4%, $p = 0.031$) on wind and ties TimesFM 3.0 (−2.5%), the same pattern as v0.8.

Table 9: CPCB test: pure PWM-X (P10) against the foundation models (relative MAE; positive means they are better). v0.8 (P8) in brackets.

Opponent	Temp	RH	Wind
Chronos-2	+3.4 L [+5.0 L]	+0.3 T [+1.4 T]	+3.2 L [+3.8 L]
TimesFM 2.5	−2.5 T [−1.0 T]	−2.3 T [−1.3 T]	+2.9 L [+3.5 L]
TimesFM 3.0	+1.0 T [+2.5 T]	+0.2 T [+1.3 T]	+3.8 L [+4.4 L]

8.3 A prospective real-time track

A live version of the pure model, trained on the six airports up to 23 September 2026 and validated to 7 November, issues a 24-hour forecast from the latest airport reports and logs it before the weather happens. Every logged hour is later scored against the reports. The first forecast was issued on 9 November 2026 at 06 UTC, so no scores exist yet. This track is the clean test that the reused test periods of this study cannot give.

9 LIMITATIONS

Test-set reuse across versions. All decisions were made on validation, but the test periods were scored after every version. The test results then shaped which ideas we tried next. For example, the

CPCB losses motivated v0.6–v0.9. Later CPCB gains are therefore exposed to adaptive overfitting to the test period. Clean evidence will come from data never scored before. To that end we have started a prospective real-time track: live airport forecasts are logged before the weather happens and scored afterwards.

Multiple comparisons. Each league involves dozens of tests. Verdicts with very small p-values, such as the WMO foundation-model wins ($p < 10^{-5}$), survive a Holm correction. Borderline ones, such as the two CPCB hybrid wind wins ($p = 0.02, 0.03$), would not survive.

One design per league. The final designs differ across leagues: v0.4's design is used at the airports and the WMO stations, and v0.8's and v0.9's on CPCB. Versions 0.6–0.9 have not yet been run as full leagues at the airports or the WMO stations.

Missing baseline. Our post-processing baseline is the bias-corrected four-model mean. A tuned gradient-boosted or neural post-processor [Rasp and Lerch 2018] trained on the same inputs would be a stronger opponent for the hybrid.

Scale and siting. The leagues are small: six airports, 20 WMO stations and 25 CPCB sensors, all in north India, with a single test period of six to nine months each. The CPCB sensors are not sited to WMO standards. Results there describe forecasts of what those sensors will read.

Opponents we could not run. GraphCast at 0.25°, FourCastNet 3 and Aurora exceeded the memory of the free T4 GPU. GenCast and WeatherNext at full resolution have no public archive. Our GenCast run is the small 1° model with one member, on 88 days.

Data dependencies. The archived global forecasts come from a third-party service [Zippenfenig 2023]. TimesFM 3.0 is used for evaluation only, under its non-commercial licence.

10 CONCLUSION

Trained only on a network's own history, a one-million-parameter station-graph model beats TimesFM 2.5, TimesFM 3.0 and Chronos-2 on every variable at the airports and the WMO stations. Used as a learned correction of global forecasts, the same model beats every operational global model at north-Indian airports when all are limited to forecasts that exist at the forecast start. The hybrid does not beat AIFS's freshest runs on airport temperature and humidity. The pure model does not beat the foundation models on wind at Delhi's rooftop sensors, or bias-corrected FourCastNet SFNO and GenCast Mini on airport temperature; the hybrid has not yet been scored against those two. The data threshold points to the remedy: at the airports, 24 months of local history remove every loss to the foundation models and 48 months turn all of them into wins, so pretraining on many more Indian stations is the natural next step. We report the full development record, including the rejected changes, so that others can judge which gains are robust.

ACKNOWLEDGMENTS

Observations: Iowa Environmental Mesonet (airport reports), NOAA Integrated Surface Database, and the Central Pollution Control Board of India (station sensors, via OpenCity). Archived operational forecasts: Open-Meteo (Previous Runs API; models by ECMWF, NOAA and DWD). Model runs: NVIDIA Earth2Studio. Foundation-model opponents: open weights of Google TimesFM 2.5 and 3.0 (3.0 under a non-commercial research licence) and Amazon Chronos-2. Use of AI tools: Claude (Anthropic) assisted with writing code, running experiments and drafting the text. Every reported number comes from the logged run output, and the authors take full responsibility for the content.

REFERENCES

Abdul Fatir Ansari et al. 2025. Chronos-2: From Univariate to Universal Forecasting. *arXiv preprint arXiv:2510.15821* (2025).

Abdul Fatir Ansari, Lorenzo Stella, Caner Turkmen, et al. 2024. Chronos: Learning the Language of Time Series. *Transactions on Machine Learning Research* (2024).

Kaifeng Bi, Lingxi Xie, Hengheng Zhang, Xin Chen, Xiaotao Gu, and Qi Tian. 2023. Accurate medium-range global weather forecasting with 3D neural networks. *Nature* 619 (2023), 533–538.

Cristian Bodnar, Wessel P. Bruinsma, Ana Lucic, Megan Stanley, Anna Allen, Johannes Brandstetter, Patrick Garvan, Maik Riechert, Jonathan A. Weyn, Haiyu Dong, et al. 2025. A Foundation Model for the Earth System. *Nature* 641, 8065 (2025). https://doi.org/10.1038/s41586-025-09005-y

Boris Bonev, Thorsten Kurth, Christian Hundt, Jaideep Pathak, Maximilian Baust, Karthik Kashinath, and Anima Anandkumar. 2023. Spherical Fourier Neural Operators: Learning Stable Dynamics on the Sphere. In *Proceedings of the 40th International Conference on Machine Learning (ICML) (PMLR, Vol. 202)*. 2806–2823.

Abhimanyu Das, Weihao Kong, Rajat Sen, and Yichen Zhou. 2024. A decoder-only foundation model for time-series forecasting. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*. arXiv:2310.10688

Francis X. Diebold and Roberto S. Mariano. 1995. Comparing Predictive Accuracy. *Journal of Business & Economic Statistics* 13, 3 (1995), 253–263.

Harry R. Glahn and Dale A. Lowry. 1972. The Use of Model Output Statistics (MOS) in Objective Weather Forecasting. *Journal of Applied Meteorology* 11, 8 (1972), 1203–1211.

Tilmann Gneiting, Adrian E. Raftery, Anton H. Westveld, and Tom Goldman. 2005. Calibrated Probabilistic Forecasting Using Ensemble Model Output Statistics and Minimum CRPS Estimation. *Monthly Weather Review* 133, 5 (2005), 1098–1118.

Roger Koenker and Gilbert Bassett. 1978. Regression Quantiles. *Econometrica* 46, 1 (1978), 33–50.

Hans R. Künsch. 1989. The Jackknife and the Bootstrap for General Stationary Observations. *The Annals of Statistics* 17, 3 (1989), 1217–1241.

Remi Lam, Alvaro Sanchez-Gonzalez, Matthew Willson, et al. 2023. Learning skillful medium-range global weather forecasting. *Science* 382, 6677 (2023), 1416–1421.

Simon Lang, Mihai Alexe, Matthew Chantry, Jesper Dramsch, Florian Pinault, Baudouin Raoult, Mariana C. A. Clare, Christian Lessig, Michael Maier-Gerber, Linus Magnusson, et al. 2024. AIFS – ECMWF's Data-Driven Forecasting System. *arXiv preprint arXiv:2406.01465* (2024).

Yaguang Li, Rose Yu, Cyrus Shahabi, and Yan Liu. 2018. Diffusion Convolutional Recurrent Neural Network: Data-Driven Traffic Forecasting. In *International Conference on Learning Representations (ICLR)*. arXiv:1707.01926

Yong Liu, Guo Qin, Zhiyuan Shi, Zhi Chen, Caiyin Yang, Xiangdong Huang, Jianmin Wang, and Mingsheng Long. 2025. Sundial: A Family of Highly Capable Time Series Foundation Models. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*. 39295–39317. arXiv:2502.00816

Jaideep Pathak, Shashank Subramanian, Peter Harrington, Sanjeev Raja, Ashesh Chattopadhyay, Morteza Mardani, Thorsten Kurth, David Hall, Zongyi Li, Kamyar Azizzadenesheli, Pedram Hassanzadeh, Karthik Kashinath, and Animashree Anandkumar. 2022. FourCastNet: A Global Data-driven High-resolution Weather Model using Adaptive Fourier Neural Operators. *arXiv preprint arXiv:2202.11214* (2022).

Ilan Price, Alvaro Sanchez-Gonzalez, Ferran Alet, Tom R. Andersson, Andrew El-Kadi, Dominic Masters, Timo Ewalds, Jacklynn Stott, Shakir Mohamed, Peter Battaglia, Remi Lam, and Matthew Willson. 2025. Probabilistic Weather Forecasting with Machine Learning. *Nature* 637, 8044 (2025), 84–90. https://doi.org/10.1038/s41586-024-08252-9

Stephan Rasp, Stephan Hoyer, Alexander Merose, Ian Langmore, et al. 2024. WeatherBench 2: A benchmark for the next generation of data-driven global weather models. *Journal of Advances in Modeling Earth Systems* (2024).

Stephan Rasp and Sebastian Lerch. 2018. Neural Networks for Postprocessing Ensemble Weather Forecasts. *Monthly Weather Review* 146, 11 (2018), 3885–3900.

Yaniv Romano, Evan Patterson, and Emmanuel Candès. 2019. Conformalized Quantile Regression. In *Advances in Neural Information Processing Systems (NeurIPS)*. arXiv:1905.03222

Zezhi Shao, Zhao Zhang, Fei Wang, Wei Wei, and Yongjun Xu. 2022. Spatial-Temporal Identity: A Simple yet Effective Baseline for Multivariate Time Series Forecasting. In *Proceedings of the 31st ACM International Conference on Information and Knowledge*

Management (CIKM). arXiv:2208.05233
Adam Smith, Neal Lott, and Russ Vose. 2011. The Integrated Surface Database: Recent Developments and Partnerships. *Bulletin of the American Meteorological Society* 92, 6 (2011), 704–708.
Gerald Woo, Chenghao Liu, Akshat Kumar, Caiming Xiong, Silvio Savarese, and Doyen Sahoo. 2024. Unified Training of Universal Time Series Forecasting Transformers. In *Proceedings of the 41st International Conference on Machine Learning.*
Zonghan Wu, Shirui Pan, Guodong Long, Jing Jiang, and Chengqi Zhang. 2019. Graph WaveNet for Deep Spatial-Temporal Graph Modeling. In *Proceedings of the 28th International Joint Conference on Artificial Intelligence (IJCAI).* arXiv:1906.00121
Patrick Zippenfenig. 2023. Open-Meteo.com Weather API. https://doi.org/10.5281/zenodo.7970649

A CANDIDATES AND TRAINING SETTINGS

Table 10: Pure-model candidates. [a]Width/layers/heads; feed-forward ratio 2. Exp.: sparse experts. Cross: half of the batches come from the partner network (airports ↔ WMO; WMO for CPCB). 14-day: day-by-day encoder. EMA: weight averaging, decay 0.998.

ID	Size[a]	Graph	Exp.	Cross	Extras	Ver.
P0	128/3/4	yes	–	–	–	v0.3
P1	192/4/6	yes	–	–	–	v0.3
P2	128/3/4	no	–	–	–	v0.3
P3	128/3/4	yes	4	–	–	v0.3
P4	128/3/4	yes	–	yes	–	v0.4
P5	192/4/6	yes	–	yes	–	v0.6
P6	128/3/4	yes	–	yes	14-day	v0.8
P7	128/3/4	yes	–	yes	EMA	v0.8
P8	128/3/4	yes	–	yes	14-day, EMA	v0.8
P9	128/3/4	yes	–	yes	PWM-X, EMA	v0.9
P10	128/3/4	yes	–	yes	PWM-X+SSL, EMA	v0.9

Hybrid designs. H1 adds the bias-corrected forecasts as inputs (v0.3 used IFS 0.25 and AIFS runs at least 48 h old; from v0.4 on, the four models' newest runs issued before the forecast start). H2 also makes their mean the base (v0.4). H3 adds cloud cover, shortwave radiation, dew point, sea-level pressure and wind direction from the same runs (v0.5). Each is trained on the league's best pure setting. Designs are compared on one seed in the order H1, H2, H3, and a design replaces the current choice only if it is better by more than 2σ.

Training.

- AdamW, learning rate 1.5×10^{-3}, weight decay 10^{-4}, batch 32 origins, 6,000 steps.
- Linear warm-up over 300 steps, then cosine decay to 10% of the peak rate.
- Validation every 500 steps; the best checkpoint is kept.
- Mixed precision on a single Kaggle GPU.
- Training origins are every hour; validation and test origins are every 6 hours.

Rolling fine-tuning (v0.7, rejected). Every 168 hours, each final model is fine-tuned for 500 steps at learning rate 3×10^{-4}, using only hours observed before that week, with half of the batches drawn from the last 720 hours.

Splits. Training, validation and test periods are given in §3. Normalisation statistics come from the training period only. On top of each league's station selection, the pipeline keeps a station only if at least 25% of its training-period temperature hours and of its test-period target hours are observed, a rule fixed before any scoring.